

to expand operations. Under standard conditions Ethereum maintained a capacity of 15 to 30 transactions per second, which is one-tenth of the transaction volume handled by centralised networks such as Visa. Network congestion occurred during peak times when market conditions changed or when popular dApps were released. The cost of transactions increased sharply while the time needed for transaction confirmation expanded from seconds to several minutes or more.

The congestion blocked blockchain technology from achieving its potential as a system that enables transparent operations without trust requirements. Enterprises faced challenges when they needed to move their main operations to new systems while developers had difficulty creating simple user interfaces.

Enter Layer-2: The express lanes built above the main road

Layer-2 solutions function as architectural enhancements that improve Layer-1 blockchains without destroying their original design. They operate like express lanes running above a main highway that has reliable yet limited capacity. The upper levels enable fast transactions that operate at minimal expenses while maintaining their connection with the verified base system underneath.

This requires computation and execution to be processed outside of the Layer-1 main chain as the secondary work is completed, and then all proof materials to return to Layer-1. This approach retains the consensus, economic security, and decentralisation of Ethereum while dramatically improving performance.

Well-designed Layer-2 solutions provide strong security protection which other Layer-1 blockchains lose when they choose to prioritise speed. They do not create additional trust requirements which must be satisfied in order to achieve final settlement. The base layer security allows users and developers to conduct thousands of transactions each second for less than one cent. But they still access security that has been tested in actual combat.

Layer-2 makes blockchain accessible to working professionals because it transforms the blockchain from an expensive experiment into a practical business infrastructure solution. In 2026, Layer-2 networks have become the default environment for most user activity, proving that scalable decentralisation is not a theoretical ideal but an engineering reality.

```
# Simple simulation: Individual vs Batch processing
import time
```

```
def process_transaction(tx):
    time.sleep(0.01) # Simulate small delay per transaction
```

```
    return f"Processed {tx}"
```

```
# Without batching (Layer-1 style)
```

```
def individual_processing(transactions):
```

```
    start = time.time()
```

```
    for tx in transactions:
```

```
        process_transaction(tx)
```

```
    return time.time() - start
```

```
# With batching (Layer-2 rollup style)
```

```
def batch_processing(transactions, batch_size=100):
```

```
    start = time.time()
```

```
    for i in range(0, len(transactions), batch_size):
```

```
        batch = transactions[i:i+batch_size]
```

```
        # In real L2: compress + submit one proof
```

```
        time.sleep(0.05) # One small delay for the whole
```

```
    batch
```

```
    return time.time() - start
```

```
txs = ["tx" + str(i) for i in range(1000)]
```

```
print("Individual time:", individual_processing(txs))
```

```
print("Batch time:", batch_processing(txs))
```

How Layer-2 solutions actually work

The core function of Layer-2 technology depends on rollup mechanisms that enable the system to combine and reduce operational data before it connects to the primary blockchain network. This is akin to a busy classroom where students work at their desks to solve problems, while the teacher collects only one summary sheet that shows the complete work of the entire class, which is verified by the main ledger.

The system executes transactions in a secure environment that maintains low costs while conducting off-chain activities, combining hundreds or thousands of transactions into one efficient package. It sends a small proof or state update to Layer-1, which gets protection from the base chain security system.

The main network experiences reduced data and processing demands because of this development. The system requires Layer-1 to check only the batch's authenticity instead of tracking each separate transaction. This results in an impressive throughput increase, which enables the system to process thousands of transactions each second while charging fees that cost pennies or even less.

The system operates through two primary types, both of which implement the same batch-and-settle method. It achieves its design goals because it maintains high security standards without adding extra demands to the base system. Layer-2 enables professionals and researchers studying distributed systems to solve performance problems through architectural separation while maintaining the essential blockchain features of transparency and immutability.